

Move to Miami: Cuban Elite Exiles, Local Economic Outcomes, and Business Formation

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Abstract

How do dispossessed business elites affect the places that receive them? We study Cuba's pre-revolutionary elite, many of whom reached Florida after 1959 with skills but little capital. We identify exiles by name, linking two 1958 Cuban elite registers to Florida telephone directories and corporate filings, and follow places where at least three elite households settled by 1964 through 1980. Matched difference-in-differences shows lower unemployment by 1980 in Miami tracts where the elite settled, but no consistent effect on employment or family income. Synthetic control estimates for 41 Florida zip codes show incorporations in elite-settlement zip codes rising from 3.81 to 10.00 times their own pre-1959 average relative to their synthetic controls in significant years, with the gap opening in the late 1960s.

Keywords: Cuban Revolution, entrepreneurship, migration, human capital, enclave economy, business formation, Cuban exiles

JEL Codes: J61, L26, N46, N86, O15

1 Introduction

Revolutions often have unintended consequences, particularly when they displace the groups that had previously managed much of a country's economic activity. Cuba was no exception. In February 1965, Fidel Castro acknowledged that ‘those who knew how to manage, those who possessed what little technical knowledge there was, were the owners, the rich’ (Castro Ruz, 1965). Many of those individuals had already left Cuba during the first years of the Revolution. This early wave of Cuban migrants, later known as the ‘Golden Exiles,’¹ was disproportionately composed of business owners, managers, professionals, and members of the upper and upper-middle classes. Their departure therefore represented a movement of people out of Cuba, but also a large transfer of entrepreneurial, managerial, and professional human capital to the United States.

We study this first wave of exiles, who left as the revolutionary government expropriated the assets they owned. Three features set this wave apart from most migrations. First, migrants usually move because the move pays, and they can go back if it does not. These exiles were pushed out by expropriation. Many saw themselves as temporary exiles and stayed in Miami to remain close to home (Thomas, 1967, p. 48), but the regime survived and going back stopped being an option. Second, migrants usually bring their savings. Many who left before the end of 1960 reached the United States with a good part of their assets. Toward the end of that year, however, the Cuban government began confiscating the real and personal property of departing refugees, and those who left afterward arrived destitute (Thomas, 1967, p. 47). Third, migrants usually join compatriots who settled before them. The nineteenth-century Cuban communities had formed in Tampa, Key West, and New York (Pérez, 1986, p. 126), and the small Cuban population in Miami before the exodus consisted of long-time residents sympathetic to the Revolution (Forment, 1989, p. 60). Later Cuban

¹The label reflects two unusual conditions at once. These migrants were exceptionally well prepared in education and occupation relative to other first-generation immigrant groups, and they were also exceptionally well received, admitted as refugees without the restrictions applied to other nationalities and supported by a federal assistance program that provided cash aid, training, and relocation services (Portes, 1969; Mitchell, 1962).

arrivals could find work in firms that earlier exiles had founded (Pérez, 1986, pp. 134–135), but the first wave arrived before those firms existed. What it brought was training, managerial experience, and the business ties its members had built in Havana.

We identify these migrants individually. Two compilations give us the names of the people we want to follow, the members of the Havana elite and the owners of Cuban businesses. The *Libro de Oro de la Sociedad Habanera* (Álvarez de Cañas et al., 1958) is an annual social register of Havana, and its 1958 edition lists 5,354 elite households covering 25,004 individuals. Its editors compiled each edition as an alphabetical list of families, with one entry per household giving the head, the spouse, the children and the children’s spouses, together with the family’s address, telephone number and, in most cases, the head’s profession. *Los propietarios de Cuba* (Jiménez Soler, 2008) is a biographical dictionary of the island’s business owners as they stood in 1958. Each of its 550 entries records the proprietor’s holdings, positions, family, last address and social clubs, and the origin of the fortune, and rates the proprietor’s economic importance on a five-point scale.² Historians of prerevolutionary Cuba have used both works before.³ We put them to a different use. Together they give us a named roster of the Cuban elite on the eve of the Revolution, and we match that roster against United States records to establish how many of these people left the island and where they ended up. To our knowledge, this is the first time either work has been transcribed into individual-level data and linked to receiving-country records. We match the names to digitized Florida telephone directories from 1959 onward (Library of Congress, 2025). This yields 1,101 matched individuals, 992 with an address precise enough to geocode and 710 that geocode successfully. We also match them to the Florida Division of Corporations bulk filing database (Florida Department of State, Division of Corporations, 2026), which yields

²We hold editions of the *Libro de Oro* for 1947 through 1960 and are transcribing them. This version of the paper uses the 1958 edition alone, the last compiled before the Revolution, so that the roster is a snapshot of the elite immediately prior to the exodus. The full series would allow entry into and exit from the register to be observed year by year.

³Roorda (2004), Guerra (2018), González Acosta (2022) and Vázquez and Pombo (2013) draw on the *Libro de Oro*, and Vázquez Fleitas (2024) tabulates the 5,791 families of its 1953 edition by neighborhood. Vázquez Fleitas (2023) draws on both the *Libro de Oro* and *Los propietarios de Cuba*. Pérez (2023) uses the companion *Registro Social de La Habana*.

1,487 distinct corporations with a matched officer. The directories tell us where the exiles settled. The corporate registry tells us what they founded.

We estimate effects on two geographies, and the split is imposed by the data rather than chosen. The first set of results, using the Mariel boatlift literature’s outcome variables as a reference, uses Census data. We use 1960 census tracts and a matched difference-in-differences estimator following Grier and Grier (2021), treating a tract as exposed when at least three elite households settle in it, which gives 46 treated tracts in the estimation sample. Then, we look at the effect of the shock on business creation. Here, the unit is the zip code and the geography is wider, covering 41 treated zip codes across Florida (31 in Miami-Dade, four in Broward, and six in Tampa), estimated by synthetic control against 102 urban Florida zip codes that received no elite household.

We find that elite settlement is associated with a reduction in unemployment when looking at the change from 1960 to 1980. We find no significant and consistent evidence of its effect on employment and median household income (a proxy for wages). Additionally, we find that settlement is associated with an increased business creation rate.

The most studied Cuban migration to the US has been that of the Mariel Boatlift, especially in economics. Card (1990), Bodvarsson et al. (2008), Borjas (2017), and Peri and Yasenov (2019) all analyze the labor market implications of this massive influx of people in such a short period of time to Miami’s economy.⁴ Yet the Mariel migrants were far less skilled than earlier exodus of business owners, professionals, and other members of Cuba’s pre-revolutionary elite. These earlier waves have been studied primarily in the sociological literature on the emergence of Miami’s Cuban enclave economy (Portes, 1987; Wilson and Portes, 1980; Doeringer and Piore, 1971; Wilson and Martin, 1982; Cobas et al., 1993; Portes and Zhou, 1996; Portes and Puhmann, 2015). We argue that the first wave of migrants has not been studied as extensively in economics due to data limitations and the lack of a research design that can adapt to these limitations. We bring the analysis of early wave

⁴Additionally, Yao et al. (2022) studies fiscal outcomes following the same migration shock

Cuban exiles to economics, with novel data and research design.

By bringing the first wave of Cuban exiles into the discussion, we contribute to the literature of elite and high-human-capital migration in economics. This literature examines how displaced elites and other highly skilled migrants affect receiving economies. This work shows that émigré scientists can transmit knowledge and stimulate innovation among us researchers (Moser et al., 2014; Ganguli, 2015), although their arrival can also reduce the productivity of incumbents working in overlapping fields and push some out of active research (Borjas and Doran, 2012). Other studies have shown that skilled refugee communities can diffuse technology and raise firm productivity over the long run (Hornung, 2014), while exiled political leaders can mobilize collective action through local newspapers and civic associations (Dippel and Heblich, 2021). More broadly, immigrant inventors and entrepreneurs contribute disproportionately to patenting, firm formation, job creation, and high-growth entrepreneurship (Akcigit et al., 2017; Azoulay et al., 2022). We contribute to this literature by identifying individuals who belong to the business elite before displacement and tracing how their arrival affected entrepreneurship and local development across receiving communities.

We also contribute to the literature on how the Cuban Revolution affected living standards. This body of work asks what Cuban living standards would have been without the Revolution (Ward and Devereux, 2012) and estimates its effects on GDP per capita and exports (Jales et al., 2018; Bastos et al., 2026), infant mortality (Geloso and Bologna Pavlik, 2021), and environmental performance (Martinez and Geloso, 2026). Devereux (2021) identifies the exit of managers and skilled workers after 1959 as one source of Cuba’s early decline. These studies measure the Revolution’s effects on the Cubans who stayed. We add to this literature by measuring its effects through the Cubans who left. The Revolution drove much of the island’s business and professional elite into exile, and we estimate what those exiles contributed to the economies that received them, measured by local unemployment and new business formation in the Florida communities where they settled.⁵

⁵The mirror question, what that outflow cost Cuba itself, remains open. Answering it would require municipal- or sector-level Cuban data for this period, which to our knowledge do not exist.

Section 2 describes how fast the Revolution took Cuba’s firms and property, who the first exiles were, why those who left after 1960 arrived with their skills but little of their wealth, and how they reassembled in Miami despite a federal program designed to resettle them elsewhere. Section 3 turns the 1958 elite registers into a named roster and links it to Florida telephone directories and corporate filings, which locate 1,101 members of the elite and 1,487 firms with an elite officer. Because the Current Population Survey used in the Mariel studies starts only in 1962, Section 4 measures labor-market outcomes with census tract data from 1950 to 1980 and business formation with yearly counts of new incorporations by zip code. Section 5 defines a tract or zip code as treated when at least three elite households settled there by 1964, before the less selected Freedom Flights began, and describes our two designs: a matched difference-in-differences for Miami census tracts and a synthetic control for Florida zip codes. Section 6 shows that by 1980 unemployment was lower in treated tracts than in matched tracts, with no consistent difference in employment or family income, and that new incorporations in treated zip codes pulled away from their synthetic controls in the late 1960s, more so where more elite households settled. Section 7 raises the treatment threshold for the tract outcomes and re-estimates business formation by synthetic difference-in-differences. Section 8 concludes.

2 The Cuban Revolution and the Golden Exiles

Cuba in the late 1950s was not a poor country. In the 1955 benchmark comparisons of Devereux (2021, pp. 18–22), Cuban output per head reached 27 percent of the United States level and consumption per head 35 percent, which placed the island among the middle-income economies of that year, alongside Argentina, Uruguay, and Puerto Rico in Latin America and Ireland, Finland, and Italy in Europe. Cuban households consumed at roughly the Irish level and above the levels recorded for Spain, Portugal, Italy, and Greece. What contemporary observers singled out was less the aggregate than the people. The 1951 World Bank mission,

known as the Truslow Report, described Cubans as “intelligent, able and quick to absorb modern knowledge” and Cuban businessmen as “shrewd and capable” (International Bank for Reconstruction and Development, 1951). That commercial and professional class is the subject of this paper, and within a decade most of it had left.

The Revolution dismantled the economy that class had operated in, and it did so quickly. Private ownership and market allocation gave way to state ownership and central planning through a sequence of expropriations and nationalizations (Domínguez, 1978; Bastos et al., 2026). The First Agrarian Reform took large agricultural estates, and measures in 1960 transferred foreign-owned oil refineries, U.S.-owned sugar mills, banks, utilities, and most large domestic industry to the state (Mesa-Lago, 1981; Jales et al., 2018). Planning was then extended through JUCEPLAN, private education nationalized, cooperatives converted into state farms, and a second Agrarian Reform in 1963 reached medium-sized holdings. Table 1 reports the state share of the means of production by sector. By 1961 the state controlled banking, wholesale and foreign trade, and education outright, along with most of industry and transportation, and by 1968 industry, construction, and retail trade had become fully state-owned as well (Mesa-Lago, 1981, p. 15). The last stage reached the smallest proprietors, when roughly 56,000 remaining nonagricultural private businesses passed to the state in the March 1968 *Ofensiva Revolucionaria*, among them restaurants, repair shops, small retailers, and street vendors (Mesa-Lago, 1981, p. 23).

The property the state took belonged to identifiable people, and the earliest departures came from among them. Those cohorts were drawn from Cuba’s propertied and professional classes and included firm owners, executives, merchants, sugar-mill owners, manufacturers, cattlemen, and established professionals (Pedraza-Bailey, 1985; García, 1996). Departures were politically ordered, with officials tied to the Batista government leaving first in late 1959, conservative opponents through 1961, and liberal opponents between 1962 and 1964 (Forment, 1989, p. 54). By the late 1960s roughly 380,000 people had gone, about one in twenty-one of Cuba’s residents (Portes, 1969, p. 505), and by 1972 the cumulative total

approached 560,000 (Bender, 1973, p. 271).

Two studies measure how far these arrivals departed from the population they were drawn from, and both reach the same answer from different data. Using records from the Cuban Refugee Emergency Center in Miami, Fagen and Brody (1964) compare refugees arriving between 1959 and September 1962 with the 1953 Cuban Census, the last complete enumeration of the Cuban occupational distribution before the Revolution. Professionals and semiprofessionals were overrepresented among refugees by more than five to one relative to the Cuban workforce, and 14.2 percent of male refugees had completed four or more years of college against 1.2 percent of Cuban adult men (Fagen and Brody, 1964, pp. 391–392).⁶ Working from a separate tabulation, Pérez (1986, p. 130) finds that up to 37 percent of household heads arriving between 1960 and 1962 had been proprietors, managers, or professionals, against fewer than 10 percent of the Cuban labor force.⁷

The composition of Cuban arrivals changed substantially over time, which is precisely why we rely on elite social registries to identify this early group and follow its members through time. The professional and managerial share of arrivals fell from 31 percent at the start of the exodus to 12 percent by 1970, by which point most arrivals were blue-collar, service, or agricultural workers (Pedraza-Bailey, 1985, pp. 11, 18). Cuban authorities also began screening departures. Castro announced on 28 September 1965 that anyone wishing to leave could do so (Bender, 1973, p. 271), and the Memorandum of Understanding that followed the resulting Camarioca boatlift established the Varadero airlift and barred the departure of men of military service age, as well as professionals, technical workers, and skilled workers whose departure would cause “a serious disturbance” in production or in the delivery of social

⁶The underlying roster covered 84,578 individuals classified by occupation and month of entry. Fagen and Brody (1964, p. 390) drop students, housewives, retired persons carrying no occupational classification, and everyone who entered before the change of regime, leaving 59,682. They caution that what remains under-represents both the earliest arrivals and the wealthier refugees, since the affluent left first and the Refugee Center did not open until early 1961.

⁷The two estimates are not measuring quite the same thing. Fagen and Brody (1964) classify individual refugees by their own occupation, while Pérez (1986) reports the occupation of household heads, which mechanically raises the professional and managerial share. That two constructions agree on the direction and rough magnitude of the selection is what makes the finding hard to attribute to either one.

services (Pedraza-Bailey, 1985, p. 16). The Freedom Flights that began in December 1965 therefore brought a larger and much less selective inflow into many of the same destinations, so a later cutoff would mix two meaningfully different populations. The contrast with 1980 is sharper still: the Mariel exodus brought more than 123,000 people whose occupational profile bore little resemblance to that of the early waves (Bach, 1980, p. 39) and who subsequently fared considerably worse in the Miami labor market (Portes and Stepick, 1985; Portes and Puhrmann, 2015).

What these migrants were permitted to take with them is what makes the shock interpretable as human capital rather than wealth. Before the end of 1960 many reached the United States holding a good part of their assets, but from that point the government began confiscating the real and personal property of departing refugees, who arrived in what Thomas (1967, p. 47) describes as a state of destitution. The rules governing removal of wealth from the island reached their stringent form by the middle of 1961 (Fagen and Brody, 1964, p. 390). Some capital did reach Miami (Wilson and Portes, 1980), and Forment (1989, p. 60) attributes it to the Batistianist wave of late 1959, which departed before any of these measures existed. For the professional and managerial arrivals who followed, what crossed the Straits was training, experience, and reputation.

The city they arrived in was not an obvious destination for a commercial class. Miami in 1959 was, in the description of Portes (1987, p. 340), a sleepy resort town at the southern tip of the country, with a labor market dominated by tourism and small industry (Wilson and Portes, 1980, p. 307). Nor was there an established Cuban community to absorb them, since the nineteenth-century Cuban settlements had formed in Tampa, Key West, and New York rather than Miami (Pérez, 1986, p. 126), and the small Cuban population already in the city consisted of long-time residents broadly sympathetic to the changes underway on the island (Forment, 1989, p. 60).

That these migrants ended up concentrated in Miami was not for want of federal effort to prevent it, and the attempt to disperse them is what makes the concentration we exploit

informative. Concerned about the pressure of arrivals on Dade County, the government established the Cuban Refugee Program in 1961, and resettlement away from South Florida became its central objective and after 1962 effectively the only alternative on offer to new arrivals (Mitchell, 1962; Portes, 1969). Between January 1961 and December 1966 the program resettled roughly 135,000 of the some 229,000 refugees it registered, placing them in nearly three thousand communities across every state (Thomas, 1967, pp. 51–52). The dispersal did not hold. A trickle-back to Miami was already being documented by the early 1970s, and by 1980 the communities that had received the largest numbers of resettled Cubans were precisely those losing Cubans to Florida (Pérez, 1986, p. 131). By 1970 more than 56 percent of the metropolitan area’s Cubans lived in the central city against 26 percent of the population as a whole (Aguirre et al., 1980, p. 39). A policy designed to scatter this population across the country instead produced, within a decade, its reassembly in one metropolitan area.

In light of this, on top of the usual labor market mechanisms argued to affect economic outcomes in receiving economies (labor supply shocks, increased demand for goods), we believe that this selected group of migrants carried with them an unusual level of human capital, a network that went into exile, and they faced language and occupational licensing frictions which would bar them from being professionals in the same manner they were in Cuba. We argue that these conditions would enable them to start businesses with more ease, as they already possessed the know-how (human capital), the same network that existed in Cuba could also form business relations in South Florida, and some might even be forced into entrepreneurship due to not being able to perform their professions in the same manner they did in prerevolutionary Cuba.

This push to entrepreneurial activity could then offer better opportunities to people arriving to the enclave, which is exactly what Wilson and Portes (1980) documents: Cubans working in enclave-owned firms received returns to education and occupational experience comparable to those of workers in the primary sector outside it, contrary to the expectation

that immigrants are confined to a low-wage secondary segment.⁸ Through these mechanisms, we should see an increase in entrepreneurial activity throughout the areas the exiles settled, and better labor markets where the enclave would form. None of the existing literature estimates what the arrival the first wave exiles did to the places they settled, which is the question we take up.

3 Matching Cuban Elites to Florida Landings

In this section, we describe the process through which we were able to get names for prerevolutionary Cuban Elites, and how we match these to existing sources to pinpoint their South Florida landings.

3.1 The 1958 Cuban elite registers

The *Libro de Oro de la Sociedad Habanera* (Álvarez de Cañas et al., 1958) is an annual social register of Havana. Its editors compiled each edition as an alphabetical list of families, and its unit of observation is the household. The 1958 edition lists the fifty noble titles then current on the island. The register also carries short biographies of prominent physicians, lawyers, and teachers, and lists the couturiers who served this clientele together with the addresses of their ateliers. Its coverage overlaps closely with the private clubs that organized elite social life, and Roorda (2004) finds roughly four fifths of the Havana Country Club’s Cuban membership listed in it.

Each record is built around a household head and spouse and then lists their children and their children’s spouses. It gives the home address, telephone number, and neighborhood, and in most cases the head’s profession and work address. The household roster multiplies

⁸The dual labor market hypothesis (Doeringer and Piore, 1971) posits a primary sector with stable jobs, returns to education, and internal mobility, and a secondary sector with low wages and limited mobility. Immigrants are typically expected to be confined to the latter. That advantage did not extend to later arrivals, and Portes and Puhrmann (2015) document pre-Mariel cohorts retaining it while later ones were largely closed off.

the names available for matching, since a daughter who married and emigrated may appear under either surname. The addresses are precise enough to place families street by street, and Vázquez Fleitas (2024) uses those in the 1953 edition to map the elite across Havana’s neighborhoods.

Three features of the register suit it to our purpose. It lists professionals and executives alongside owners of capital, which is the human capital we want to follow. Because it centers on Havana, it covers the city where Cuba’s commercial, financial, and professional class was concentrated. And because it appeared annually, each edition fixes the roster to a date, so the 1958 volume captures the elite as it stood on the eve of the Revolution.

Los propietarios de Cuba (Jiménez Soler, 2008) adds the economic dimension. It is a biographical dictionary of the island’s largest business proprietors as they stood in 1958. Each of its 550 entries records the proprietor’s holdings, positions, family, last address and social clubs, nationality, profession, and the origin of the fortune, and rates the proprietor’s economic importance on a five-point scale. The two sources describe overlapping populations, and together they cover both the social and the economic elite.

Historians of prerevolutionary Cuba have used both works.⁹ We use them as a sampling frame. Together they name the people whose departure we trace, and those names are what we carry into the United States records described next.¹⁰

3.1.1 White pages: locating the elite in Florida

Both linkages share a name model. Names are stripped of accents, honorifics and punctuation, then split into a first given name, a paternal surname and a maternal surname. A person

⁹Roorda (2004) uses the *Libro de Oro* to map the membership of Havana’s clubs and Guerra (2018) to identify individuals. González Acosta (2022) draws on the 1958 edition to trace surviving noble titles, and Vázquez and Pombo (2013) draws on the register to reconstruct the couture trade. Vázquez Fleitas (2024) tabulates the 1953 edition by neighborhood. Vázquez Fleitas (2023) draws on both the *Libro de Oro* and *Los propietarios de Cuba*, and reads the *Libro de Oro* for 1947 through 1958 against the *Registro Social de La Habana* for 1950 through 1958 to observe turnover in the roster. Pérez (2023) follows one family across the Revolution through the *Registro Social*.

¹⁰We hold editions of the *Libro de Oro* for 1947 through 1960. We use 1958 because it is the last edition compiled before the Revolution. The 1953 edition contains 5,791 families (Vázquez Fleitas, 2024), so the register is stable in scale across the decade. The full series would let entry and exit be observed year by year.

recorded in Havana as “Márquez Sterling, Carlos” may appear in a U.S. source filed compound (*Márquez Sterling Carlos*) or with paternal-only (*Márquez Carlos*).

To locate where elites lived after moving out of Cuba, we use the Library of Congress collection of digitized telephone directories (Library of Congress, 2025). Here, we were able to find directories from 1959 onward. The listing format for individuals looks like the following:

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Alvarez Enrique 20141 NW 15 Av 624-0296
Alvarez Argudin Mario 211 Antilla AvCG--445-3207
Alvarez Aramis Jr 3900 Gardn Av MB---JE 4-2055
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In total we match 1,101 individuals, of whom 992 carry an address precise enough to geocode and 710 return coordinates. Acceptance of a link depends on certain rules: if the link uses both paternal and maternal last names, it is accepted right away; if it uses the paternal last name alone, we require the given name in full and drop last names appearing more than twelve times in that directory volume. This rule is a “safety” as very common last names could create spurious matches, like García or Rodríguez. A match on the maternal last name alone is never accepted, and where a single listing matches more than one elite person we drop the ambiguous paternal-only matches. Out of all links, 91.2% are paternal only, and only 8.8% are compound. We locate each elite landing in a 1960 census tract.

Figure 1 shows where elites established their residence upon arrival to the Miami area. Every point is a distinct elite household, and the shading indicates the 1960 Spanish-surname population share. Most of the households fall outside the tracts with the largest hispanic-shares: They were part of the formation of the Cuban enclave.

Table 2 shows the mix of occupations of individuals we could match in Florida, compared to the mix of occupations of all the households in the Havana elite database. It shows that no occupational group from the pre-revolution era is under or over-represented in our matched database. There was no apparent selection of any specific occupation into migrating to Florida after the revolution. This is the data we use to establish treatment in our empirical tests.

3.1.2 Florida corporate filings: locating their firms

We can track where and when they opened up businesses in exile using the Florida Division of Corporations (Sunbiz) bulk filing database (Florida Department of State, Division of Corporations, 2026). This dataset contains all business entities ever registered in Florida, going back to the early 1900s. For each firm, the data records the corporation name, officer fields, filing date, status, and last-event date.

The linkage here matches the same elite names against the officer fields of the Florida corporate registry, and it is what identifies elite-owned *firms* as opposed to elite *residences*. The registry is a fixed-width file. We read the document number, entity name, filing year and principal address, together with up to seven officer name slots per corporation, each parsed as a surname and a given name.

The procedure mirrors the directory matcher, with two differences. First, surname frequency is computed once across the registry’s officer fields rather than within a single volume, and the acceptance cap is looser, as we have a bigger volume of names involved: a paternal-only match requires the full given name and a surname occurring no more than twenty-five times. Second, the unit is the corporation. After matching, records are deduplicated on the document number, so a firm with two matched officers is counted once. This procedure yields 1,487 matched corporations.

Figure 2 shows the businesses we could match to distinct elite members that we could locate. The timing shows that a small number of businesses are founded in the 1960’s, specially along 8th street (*Calle Ocho*), and West Flagler. In the following decade there is a major expansion of Cuban Elite businesses, and major clusters form along Little Havana, Coral Gables, Downtown, and Brickell. Finally, Figure 3 shows where these people lived, and where they opened up businesses.

This dataset’s matches are not used for anything but the figures included in this section, as treatment is solely based on the settlement of elites, rather than on where they opened up businesses. We use the Sunbiz data to study the effect of elite settlement on new business

formation.

4 Data

4.1 Census Data

In our initial results, we examine some labor market outcomes that the Mariel literature has analyzed before—wages, unemployment, and employment—but the CPS microdata which is the one that Card (1990), Borjas (2017), and Peri and Yasenov (2019) use is only available starting 1962, which postdates our initial treatment. As a substitute, we use the National Historical Geographic Information System, which provides tract-level tabulations of the 1950, 1960, 1970, and 1980 censuses (Manson et al., 2026). Tract boundaries change between censuses, so we harmonize every year to 1960 boundaries by apportioning each earlier or later tract’s counts according to the share of its land area falling in each 1960 tract, following Lee and Lin (2018, p.672).¹¹

The 1950 census binds everything that follows, as our design matches on 1950 characteristics. This also has a consequence worth stating plainly: no Florida county except Dade was tracted in 1950, so any specification built on 1950 characteristics confines the within-state comparison to Miami itself. We report no Florida-metros control pool because the census does not permit one.

We draw controls from three pools, holding the treated set fixed: the rest of Dade County, the census South, and the whole country. A tract that received one or two elite households is excluded from every pool rather than used as a control. This is because leaving them among the controls would bias the comparison towards zero, and the zip-code design applies the same rule. Control counts vary slightly across outcomes and horizons, since a tract enters an

¹¹Unlike the Longitudinal Tract Database (Logan et al., 2014), which uses block-population weights for 1970 onward, we use land area across all decades because block populations are unavailable for 1950. We recompute rates and shares from the apportioned counts; for median variables, we calculate a weighted average across overlapping tracts using the intersection area multiplied by population as the weight. Only 1960 values are observed directly.

outcome’s estimation only where that outcome is observed. The pools range from 41 to 47 control tracts within Dade, 1,750 to 1,759 in the South, and 9,277 to 9,324 nationally, against 46 treated tracts throughout.

We estimate effects on three outcomes: the unemployment rate, the employment-to-population ratio, and log median family income. These are the outcomes of the Mariel literature, whose headline variable is a wage (Card, 1990); no tract-level wage exists for 1960, so median family income is the closest analog we can find in this setting. Matching uses five 1960 characteristics common to every outcome: log median family income, median school years, the Spanish-surname share, the professional and managerial share, and log population density, together with the outcome’s own 1950 and 1960 levels. This gives six or seven covariates per outcome, six where the outcome is itself one of the five. Of the 177 Dade tracts in the harmonized panel, only 142 exist in 1950, and 128 have the covariates the specification requires. Excluding the 35 tracts that received one or two elite households leaves 93, of which 46 are treated: all 46 of the 54 treated tracts that survive into the covariate sample. Table 3 shows the summary statistics for the matching covariates and 1950 and 1960 outcome levels for treated tracts and the control pools.

4.2 Florida Corporate Filings

Business formation comes from the Florida Division of Corporations bulk filing database (Florida Department of State, Division of Corporations, 2026), which records every entity ever registered in the state. We parse 668,468 filings with filing years between 1940 and 1980. Of these, 398,740 carry a valid Florida zip code in the registry, and we recover a zip code for 57,875 more by geocoding their street address, for a total of 456,615, or 68.3 percent, spread across 1,896 distinct zips.¹²

¹²ZIP codes were introduced in 1963, so only firms that had activity in their filings after this date have a ZIP code in their address. On top of that, some firms from after 1963 do not have a ZIP code, making the count of firms per Zip lower than the real one. To solve this, we recovered – with the help of the US Census Geocoder – 57,875 from the 70,022 filings that have a full address but no ZIP. This implies an increase of 33% in pretreatment filing records. There is an extra 149,000 filings without any street address, which we are

This series is an aggregate count over the entire registry. It counts every incorporation in a zip-year, whatever the names of the officers, and it is a different object from the 1,487 elite-linked corporations of Section 3.1.2, which are name-matched. Treatment enters through the zip codes in which the elite settled, not through the firms they founded. The distinction matters for what the estimates mean: a name-matched outcome would measure only what the exiles built themselves, while an aggregate count measures whether the places they settled became places where everyone incorporated.

5 Empirical Strategy

5.1 Defining treatment

We use settlement as treatment: the elite persons linked to post-1959 Florida directories, linked to a street address, and dated by the year in which they first appear. We treat that first appearance as the arrival date, and we count only those appearing no later than 1964.

For the census outcomes the unit is the 1960 census tract. Each landing is placed inside a tract containing it in the Miami area, which locates 414 landings. A tract is treated if at least three elite households settle in it. Of the 177 Dade tracts in the panel, 97 receive at least one landing and 54 receive at least three, and 46 of those 54 survive into the estimation sample.

For business formation the unit is the zip code, and here the geography is wider. We recover a Florida zip code from the address of 504 landings, and treat a zip that receives at least three elite households, the same cut the tract design uses. This yields 41 treated zips out of the 79 that receive any elite settlement at all, and those 41 contain 454 of the 504 landings.

The two designs differ in scope: the tract analysis is confined to Miami (because in 1950 census data it is the only metro area available in Florida), while the treated zips include 31 in Miami-Dade, four in Broward, and six in Tampa. The business-formation results describe

not able to recover.

the places in Florida where the elite settled, not Miami alone.

We set the threshold at three households instead of one, because a single household is a weak signal of settlement and is more exposed to a false match in the linkage, while three establishes a presence that could plausibly matter for a neighborhood. It can also serve as a signal that these neighborhoods were the ones where newly arrived Cubans would later settle in, after the first arrivals had laid the foundations.

The 1964 cutoff isolates the early elite wave. The Freedom Flights that began in 1965 brought a far larger and much less selected Cuban inflow into the same places, and a later cutoff would mix the two. Now we introduce our two empirical methods.

5.2 Matched difference-in-differences

Our estimator is a difference in differences, implemented by matching. For each outcome we ask how much more treated tracts changed between 1960 and the endpoint than tracts that resembled them beforehand. The outcome is therefore a change rather than a level, and the estimate is the contrast between that change in treated tracts and the same change in a weighted set of matched controls. The nine estimators described below differ only in how they choose and weight those controls.

Here, the two steps do different work, and both are necessary. Matching balances the pre-treatment characteristics we observe for the census tracts, while differencing removes unobserved tract characteristics that do not change over time (An and Winship, 2017). We follow Grier and Grier (2021) and Bologna Pavlik and Young (2026) in matching treated units to controls and estimating the effect on a differenced outcome.¹³

We match on the outcome’s own 1950 and 1960 levels together with five 1960 structural characteristics: log median family income, median school years, the Spanish-surname share,

¹³Our implementation follows the public replication code of Bologna Pavlik and Young (2026), using `psmatch2` (Leuven and Sianesi, 2003) and the Mahalanobis estimators with Stata’s `teffects nnmatch`, but we follow Grier and Grier (2021) by using the option `common` so treated units that do not have any comparable controls are dropped in the PSM specifications. This removes 16 to 21 of the 46 treated tracts when using the Dade pool and none for the South or national pools in the PSM estimations.

the professional and managerial share, and log population density. Two pre-treatment periods are the most the data allow. The census is decennial, and 1950 is the first year in which Miami was tracted at all, so no earlier tract observation exists for any of our units.

We report nine matching estimators: propensity-score matching at one to four nearest neighbours and with a normal kernel, and Mahalanobis matching at one through four nearest neighbours. Inference differs by family: We bootstrap the average effect on the treated with 250 replications for the propensity-score estimators, and use the bias adjustment of Abadie and Imbens (2011) with the robust variance of Abadie and Imbens (2006) for the Mahalanobis estimators. x

5.3 Synthetic control for business formation

We estimate the effect of elite settlement on business formation by synthetic control (Abadie et al., 2010). Each of the 41 treated zip codes – those receiving at least three elite households – is matched to a weighted combination of 102 donor zip codes, using annual counts from 1946 to 1980 with treatment beginning in 1959. Zip codes that receive one or two elite households are dropped rather than used as donors, following the same logic as the tract design.

The outcome is each zip code’s incorporations divided by its own 1946–58 mean. This means that the effect should be read as a multiple of its own pre-treatment average, and that every unit averages one before treatment. We do this normalization because a good part of all of our treated zips sit above all the control zips. By normalizing we remove the difference between the donors and the treated zips, and instead of trying to create a counterfactual based on the levels of the outcome, we create a counterfactual that follows proportional movements. In Figure 7 we plot the trajectories of treated and donor zips before and after doing the normalization.

Since fitting a synthetic control on 1,817 donors can lead to interpolation bias (Abadie, 2021, p. 401) due to many dissimilar units, we apply a restriction to the donor pool. Of the Florida zip codes that appear in the filings and receive no elite household, we keep

the 102 with at least 26 filings between 1946 and 1958, an average of two a year. This restriction operates as a floor, discarding zips that have too little business activity to be comparable to our treated zips. This matters because of normalization: a zip code with a very low pretreatment average might seem like it “boomed” with a much smaller increase in the outcome than other zips. A zip with a one firm per year pre-treatment average that had twenty after treatment would read as an increase of twenty times, so if these are included in the estimation, they would bias our estimates towards zero or below.¹⁴

We use all thirteen pre-treatment years as predictors, following Ferman et al. (2020), as we do not have zip-level covariates for this period. Estimations are done using `allsynth` (Wiltshire, 2022): a separate fit for each treated zip code against a common 1959 treatment date, averaged across the 41 as in Cavallo et al. (2013), with the gap bias-corrected by ridge regression (Ben-Michael et al., 2021; Abadie and L’Hour, 2021) as in Bastos et al. (2026).

For inference we use placebo-based tests. We follow Abadie et al. (2010) by fitting a synthetic control for each of the 102 donors against the rest of the pool. Since we have 41 treated zips, we also follow Cavallo et al. (2013) by averaging bundles of 41 donors. We then take 100,000 bundles, with donors chosen at random with replacement to get our placebos. For the overall p-value, we use the share of bundles whose post- to pre-treatment RMSPE ratio is larger than the treated average’s; and for the per-year p-values we divide each year’s gap by its own pre-treatment RMSPE for the treated and every bundle, and use the share of bundles whose scaled gap is larger in absolute value.

¹⁴Including all donors with enough data – every zero-elite zip code with at least two filings between 1946 and 1958, 699 in total – leads to an estimated effect of -9.32 , confirming our suspicion.

6 Results

6.1 Economic outcomes

In our first batch of results, we study unemployment, employment, and median family income. We report the results as figures, and the full tables with all estimators are in Tables A1–A6 in the appendix. Figures show the Average Treatment Effect on the Treated on the change in the outcome variables from 1960 to 1970 and from 1960 to 1980, divided into three different estimates based on the control pool.

The results for the change in unemployment can be seen in Figure 4. First, from 1960 to 1970, we see that when compared to other Dade tracts the effect is significant in four out of nine estimators, with all estimates being positive, and when being compared to tracts in the south and countrywide, there is a statistically significant negative effect on the treated tracts, with the national tracts comparison being so across the board. Then, for the change from 1960 to 1980, all three control pool specifications show a negative effect that is statistically significant in most estimators.

This means that there is some evidence that in the first years after settlement, treated tracts had a higher unemployment rate than comparable Dade county controls, but still had a lower rate than tracts in a wider pool, and by the eighties it had a lower unemployment rate than all comparable tracts across control groups. During the first period, migrants might have had a hard time integrating into the labor market, facing a higher or similar unemployment rate than their peers in Dade county, but by 1980 people settled in treated tracts had significantly better employment outcomes.

Then, we turn to Figures 5 and 6 for employment and median family income. We see some statistical significance when looking at the changes in the shorter period, but very few significant effects in the longer period. Employment shows a significant positive effect on the treated when compared to tracts in the widest set of controls – nationally – and family income does too when compared to both the Southern and the US tracts. These results

would indicate that in the longer term, treated tracts do not differ in income and employment levels, although there is some evidence that in the shorter time period they had higher levels than some of their peers.

Taken together, this is not a perfect replication of the design and outcomes that have been studied in the later period of Mariel arrivals, but we believe it is the closest we can get to try take their studies back to the first wave of Cuban exile arrivals. Having lower unemployment rates is consistent with what the sociological literature has found, where the newly formed enclave economy was offering opportunities to newer migrants joining, which are not found in other migrant communities throughout the Miami economy.

6.2 Business formation

Now we turn to our results on business formation. These are shown in Figure 8. Before treatment both the treated zip average and the synthetic control are relatively flat and follow each other's trajectory closely. Following treatment in 1959, the treated zips fall slightly behind their synthetic control, showing a significant negative effect at the 10% level in 1959, 1960 and 1962 through 1964 (at 5% in 1962–64). The effect turns positive in 1968 and is significant in every year from 1973 to 1980, at 5% in all but 1976. Because the outcome has been normalized, the result is not in terms of newly formed businesses, but in terms of the pre-treatment average. For the positive significant years, the magnitude ranges from 3.81 to 10.00 times the pretreatment average, which with a pretreatment average of 9.98 new firms per zip per year, would mean the effect ranges from 38 to 100 additional incorporations per zip code per year relative to the synthetic counterfactual. These results mean that for the zips where the Cuban elites settled, there is a positive and significant effect on business formation, especially after 1973.

The lag in these estimates is itself informative. Business formation does not respond immediately, and the gap only becomes positive after 1968, which is what would happen if the capital these migrants held had to be rebuilt before it could be deployed. Professional

practice in the United States required licenses that Cuban credentials did not confer, and only about a quarter of arrivals had enough English to hold a job (Mitchell, 1962). Founding a firm required neither, and it is the use of accumulated commercial knowledge least obstructed by the institutions of the receiving country.

To further augment our empirical design, we also do a dose ladder, to see how the effect changes when we increase the treatment thresholds. In Figure 9 we show the synthetic control gaps for a set of different thresholds starting at 3 landings per zip (original specification) up to 20 landings (which yields only seven treated units). As the figure suggests, the effect of elite landings on business formation rises with the dose, though not at every step. The respective ATT's and p-values are shown on the right panel, with the largest ATT being that of the largest dose (20 elite landings per zip).

The ATT of the 20-landings per zip specification is 7.82 times its own pretreatment average, significant at the 1% level. This last result is comprised of only seven treated units, which represent the heart of what would become the core of the Cuban enclave in Southern Florida: Little Havana, Coral Gables, Brickell / east Little Havana, Little Havana west of downtown, Coral Way, Westchester, and Edgewater.

7 Robustness

7.1 Increasing dose for neighborhood outcomes

Since we see the effect of elite landings grow as the dosage increases for business formation, we also ran the neighborhood outcomes with different levels of doses. Timely, we look at how the effect changes if the treatment threshold is increased to five and seven households per tract. Instead of reporting figures, we report a summary table (Table A7) which shows the proportion of the estimations that are significant, as well as the median estimated effect.

There are many instances where the proportion of significant estimates changes as we increase doses, like Unemployment rate-South-1970, which gains and loses significance with

5 and 7 households respectively. Then, other results keep their levels of significance, with some changes in magnitude, like the median effect of unemployment going from -0.027 with 3 households to -0.044 with 7 households in the national pool. We view this as evidence that our initial specification – using 3 households as a minimum for treatment of a tract – is a strong enough signal, where we are able to see the headline result of the matching strategy: a significant, although small, negative effect of cuban elite settlement on unemployment rate in treated tracts.

7.2 Synthetic DiD for business formation

As an alternative estimation method, we use a Synthetic Differences in Differences method, developed by Arkhangelsky et al. (2021), using the `sdid` Stata package by Clarke et al. (2023). Under this estimator, using the same treatment threshold as the original synth, and the exact same pool of donors, we see an ATT of 30.49 firms per zip per year (Table A8 and Figure A1), significant at the 1% level using bootstrap inference techniques. This effect is of a bigger magnitude compared to the one we find in the normalized specification, but it confirms the same trend we found in the main specification: the first years after treatment do not show any significance, but starting 1965 the results become significant and positive. This lends evidence for the accuracy of our main estimator.

8 Discussion and Conclusion

Since the Cuban Revolution in 1959, roughly one-fifth of Cuba’s native population has left the island, with many more wanting to but lacking the means to do so, constrained by geographic isolation. We focus on the first people who left the island after Fidel Castro’s regime started expropriating property, especially that of the business and professional class. Many of those who left after 1960 arrived with little more than what they could carry in their luggage. With little capital, they had to start their lives from scratch after having lived comfortably

back home. These people would lay the foundations for an economic enclave that is now an important arm of South Florida’s economy.

There is economic literature on Cuban migration, but it focuses mostly on the Mariel boatlift of 1980. The Mariel boatlift caused a clean, short, and sudden shock to the receiving economy in Southern Florida, making it ideal for studying the economic effects of migration. We look back in time, focus on the first wave of migrants from Cuba to Florida following the revolution, and provide the first evidence of how this massive movement of people affected the places where it settled.

We create a research design aiming at extending the same Mariel-style analyses to an earlier wave. We do not see consistent differences in income or employment between the areas where exiles settled and the control areas, but we do find that treated areas experienced a more favorable change in unemployment than comparable controls. Data show that between 1960 and 1980 the average unemployment rate rose by about three percentage points in the Southern and national control tracts, whereas it stayed roughly flat in treated tracts. After matching treated areas to the most similar controls we can find, the 1960–80 change in unemployment remains 2.0 to 2.7 percentage points lower in treated tracts.

Additionally, to our knowledge we are the first study to use the Florida Corporate Filings (Sunbiz) database to analyze the effect of a sudden migration shock on business formation. Using the synthetic control method, we find that incorporations in zip codes where Cuban elite migrants settled pulled away from their synthetic controls in the late 1960s. The average post-1959 effect is 2.15 times each zip code’s own pre-1959 level, and it grows with the number of elite households, reaching 7.8 times in the seven zip codes that became the core of Miami’s Cuban enclave. In levels, the synthetic difference-in-differences puts the effect at about 30 additional incorporations per zip code per year.

The timing of these results fits the mechanisms we describe in Section 2. The exiles carried the know-how of having run firms and practices in Havana, along with a network of family and social ties that accompanied them into exile. Many could not return to their

former occupations, since professional licenses did not transfer, and only about a quarter of arrivals spoke enough English to hold a job (Mitchell, 1962). Founding a firm required neither, which could push them into business once they had rebuilt some of the capital a firm needs. That delay is consistent with incorporations pulling ahead only in the late 1960s. Firms founded by exiles could then hire later Cuban arrivals, and Wilson and Portes (1980) find that Cubans working in Cuban-owned firms received returns to education and experience comparable to those of workers outside the enclave. This is consistent with lower unemployment by 1980, when the elite settled. Our estimates do not yet separate these channels, and they describe places rather than the exiles alone. The Freedom Flights brought some 260,500 Cubans between 1965 and 1973 (Pérez, 1986, p. 130), and our outcomes count every incorporation and every resident, so the effects include the arrivals, customers, and firms that elite settlement drew.

Our results indicate that places where members of the pre-revolutionary Cuban elite settled experienced higher business formation and, by 1980, lower unemployment than comparable places. The business formation effect appeared only in the late 1960s, about a decade after the first arrivals. Two implications follow. First, evaluations of skilled refugee inflows that rely on short post-arrival windows, or on wage and income outcomes alone, may understate their effects on receiving economies, since in this setting the effects appeared in firm formation and with a long lag. Second, the results are consistent with human capital and social ties generating local business activity even when migrants arrive with little financial capital. Distinguishing these channels from the demand and labor supply of later Cuban arrivals requires classifying new firms by their officers' identities, which Florida corporate filings make possible.

9 Limitations and Next Steps

The most prominent limitation of this study is the fact that our matching procedure to the US records does not tell us if the people we see located in Florida are the actual people. We impose some filters to rule out spurious matches based on common hispanic and Cuban names but are not 100% sure the matches are accurate. We are working on trying to find ways to improve the accuracy of our matches.

As for things to do further down the path, the main thing we are looking at is using the microdata from the 1940 and 1950 censuses to create a “shift-share” instrument. Although the enclave was formed where there was no previous Cuban presence or formed community, it could prove to be useful. Additionally, we are exploring ways to parse out the network and the human capital effects.

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Tables

Table 1: State Share of the Means of Production and Services, 1961–1968

Sector	1961	1963	1968
Agriculture	37	70	70
Industry	85	95	100
Construction	80	98	100
Transportation	92	95	98
Retail trade	52	75	100
Wholesale & foreign trade	100	100	100
Banking	100	100	100
Education	100	100	100

Notes: Entries report the share of the means of production under state ownership in each sector, in percent. Entries refer to ownership rather than to the state's share of production. Source: Mesa-Lago (1981, p. 15).

Table 2: Occupations of the Havana elite and of the exiles matched to Florida, 1958

Occupation of household head	Havana elite		Exiles		Diff.
	<i>n</i>	%	<i>n</i>	%	
Business owner or executive	547	16.3	100	18.8	+2.5
Agriculture and sugar	91	2.7	19	3.6	+0.9
Law	753	22.5	112	21.1	−1.4
Medicine	651	19.4	97	18.3	−1.2
Engineering and architecture	304	9.1	49	9.2	+0.2
Accounting and finance	111	3.3	28	5.3	+2.0
Other professional	71	2.1	10	1.9	−0.2
Politics and military	70	2.1	13	2.4	+0.4
Property only, no stated occupation	327	9.8	46	8.7	−1.1
Other / not classifiable	424	12.7	57	10.7	−1.9
Households with an occupation recorded	3,349	62.6	531	63.7	
Households in source	5,354		833		

Occupational composition of every household in the *Libro de Oro de la Sociedad Habanera* (1958), the register of the pre-revolutionary Havana elite, beside the subset whose members are matched to post-1959 Florida telephone directories by the record-linkage procedure. “Exiles” are the matched households. Percentages are of households with an occupation recorded (3,349 and 531); the final column is the exile share minus the Havana share, in percentage points.

Table 3: Summary statistics, census tracts

	Treated <i>N</i> = 46		Dade pool <i>N</i> = 47		South pool <i>N</i> = 1,758		National pool <i>N</i> = 9,320	
	Mean	S.D.	Mean	S.D.	Mean	S.D.	Mean	S.D.
<i>Matching covariates, 1960</i>								
Log median family income	8.533	0.211	8.589	0.315	8.600	0.381	8.771	0.295
Median school years	11.170	1.083	10.932	1.697	10.374	1.977	10.851	1.626
Spanish-surname share	0.020	0.027	0.006	0.007	0.017	0.072	0.025	0.087
Professional and managerial share	0.231	0.096	0.226	0.140	0.200	0.140	0.205	0.125
Log population density	7.967	0.339	7.420	1.104	7.494	1.389	7.831	1.243
<i>Outcome levels, 1950</i>								
Unemployment rate	0.060	0.019	0.050	0.017	0.038	0.025	0.050	0.043
Employment/population	0.404	0.036	0.410	0.189	0.403	0.070	0.406	0.626
Log median family income	7.978	0.190	7.969	0.248	7.977	0.345	8.117	0.275
<i>Outcome levels, 1960</i>								
Unemployment rate	0.060	0.021	0.052	0.027	0.046	0.030	0.051	0.035
Employment/population	0.420	0.036	0.385	0.036	0.392	0.080	0.393	0.072
Log median family income	8.533	0.211	8.589	0.315	8.600	0.381	8.771	0.295
<i>Outcome changes, 1960–70</i>								
Unemployment rate	-0.019	0.018	-0.018	0.022	-0.005	0.026	-0.002	0.028
Employment/population	0.020	0.057	0.032	0.067	0.013	0.064	0.009	0.061
Log median family income	0.465	0.091	0.509	0.142	0.447	0.211	0.435	0.155
<i>Outcome changes, 1960–80</i>								
Unemployment rate	-0.001	0.020	0.002	0.022	0.026	0.042	0.032	0.047
Employment/population	0.031	0.075	0.009	0.167	0.036	0.108	0.035	0.102
Log median family income	1.080	0.153	1.107	0.237	1.091	0.281	1.048	0.234

Treated tracts are the 46 Dade County tracts receiving at least three elite households. Control pools are untreated tracts in the rest of Dade County, the census South and the whole country; the Dade and South pools are subsets of the national one. Tracts receiving one or two elite households are excluded from every pool. All tracts have the five 1960 matching covariates; *N* is the number of tracts, and a few outcome levels are unreported. Every matching estimator uses the five covariates together with the outcome's own 1950 and 1960 levels, and estimates the effect on the 1960–70 and 1960–80 changes. Values from 1950, 1970 and 1980 are harmonized to 1960 tract boundaries.

Figures

Cuban elite landings vs. 1960 Hispanic geography, Greater Miami

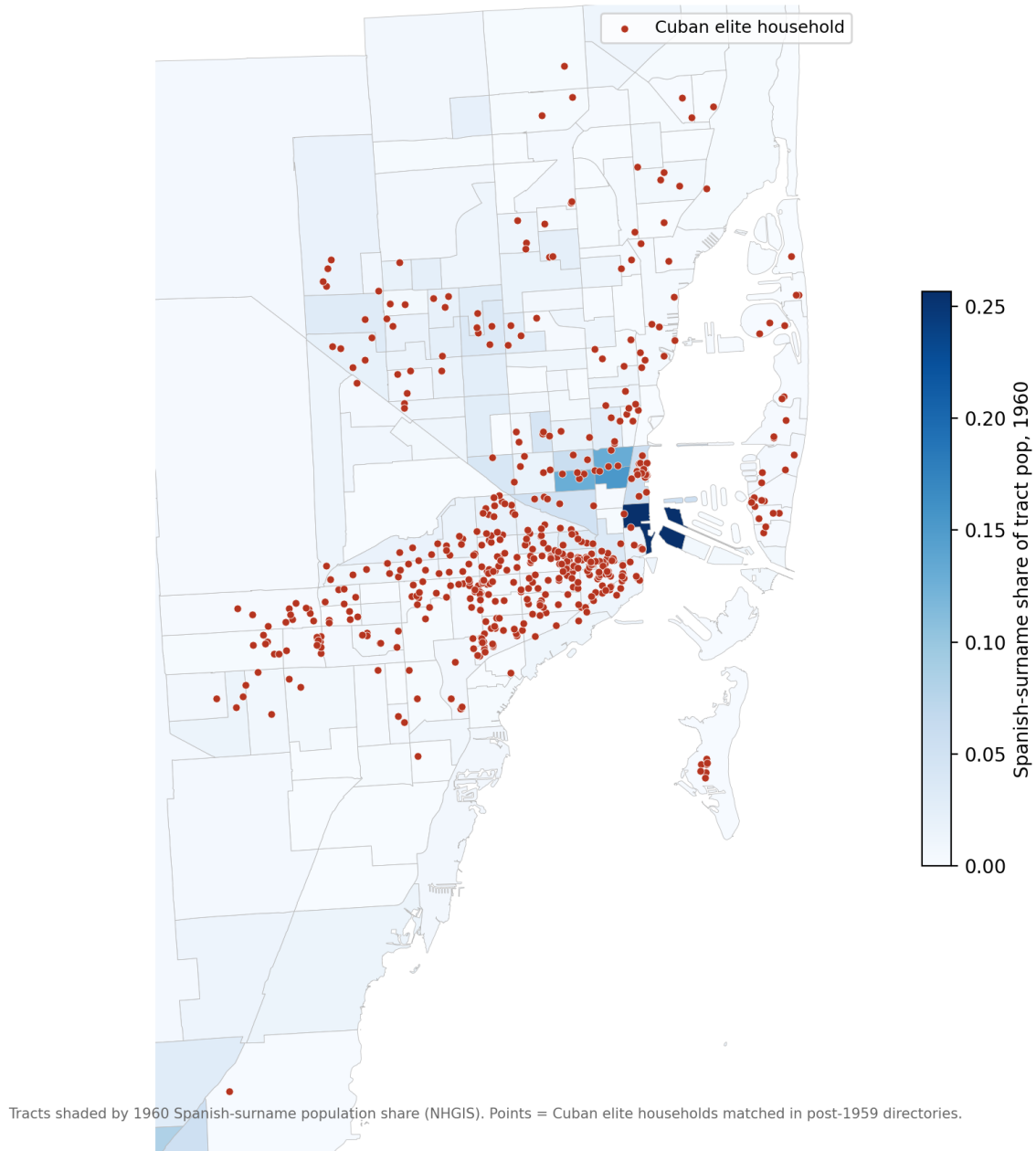


Figure 1: Elite settlement and the pre-existing Hispanic geography of Miami. Elite households located from the 1959–64 phone directories, aggregated to 1960 census tracts, against the 1960 Spanish-surname population share.

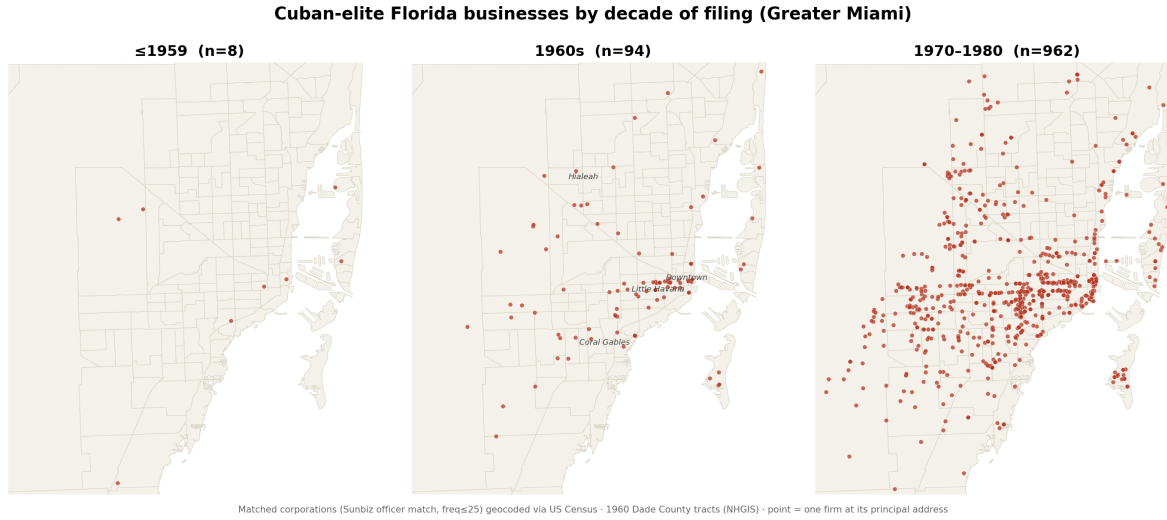


Figure 2: Businesses owned by the exiled elite, by decade of first appearance. Firms are matched from the Florida corporate registry and the city directories to the linked elite roster.

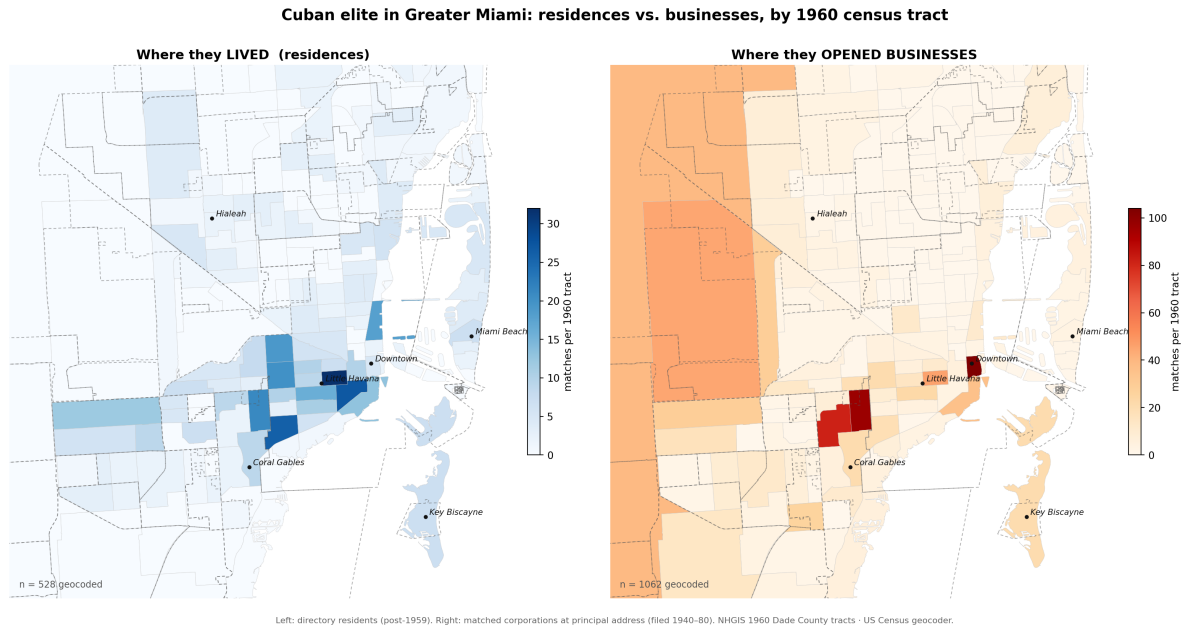


Figure 3: Residential versus business geography of the elite, by 1960 census tract. Left: elite households. Right: elite-linked business addresses. The two distributions diverge — residence and commerce did not collocate.

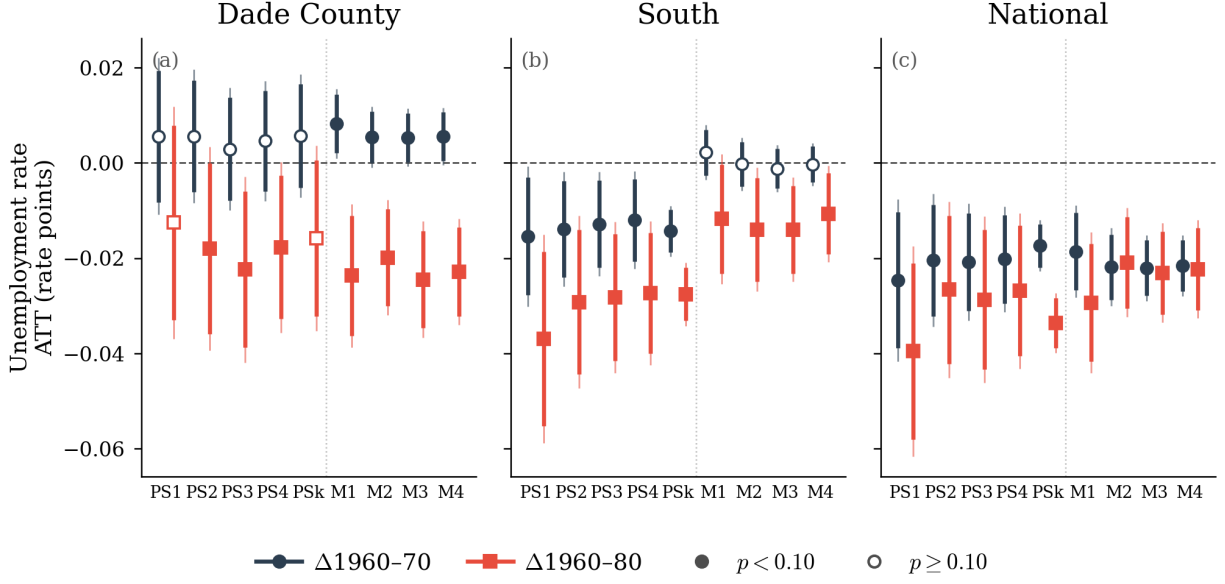


Figure 4: Unemployment rate: every matching estimator, by control pool. Estimates are average effects on the treated for the 1960–70 and 1960–80 changes. Marker fill indicates $p < 0.10$; thick whiskers are 90% confidence intervals and thin whiskers 95%. PS1–PS4 and PSk are `psmatch2` with a logit propensity score and common support (one to four nearest neighbours and a kernel), bootstrapped 250 replications; M1–M4 are `teffects nnmatch` on the Mahalanobis metric with Abadie–Imbens bias adjustment. No estimate is selected on significance. Treatment is ≥ 3 elite households; tracts holding one or two are dropped rather than used as controls.

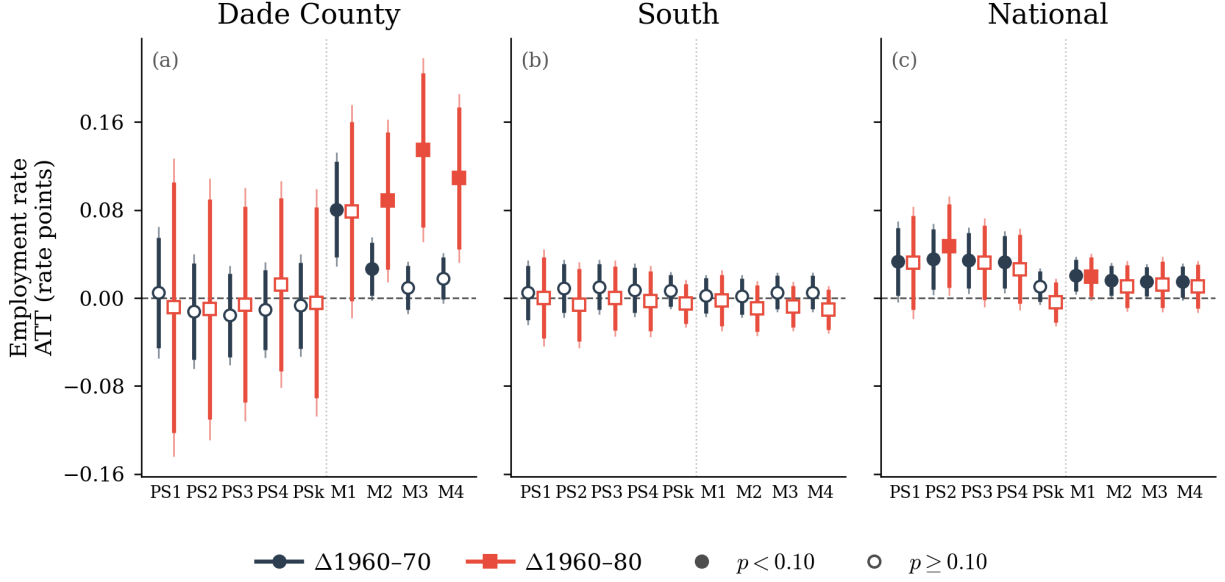


Figure 5: Employment rate: every matching estimator, by control pool. Estimates are average effects on the treated for the 1960–70 and 1960–80 changes. Marker fill indicates $p < 0.10$; thick whiskers are 90% confidence intervals and thin whiskers 95%. PS1–PS4 and PSk are `psmatch2` with a logit propensity score and common support (one to four nearest neighbours and a kernel), bootstrapped 250 replications; M1–M4 are `teffects nnmatch` on the Mahalanobis metric with Abadie–Imbens bias adjustment. No estimate is selected on significance. Treatment is ≥ 3 elite households; tracts holding one or two are dropped rather than used as controls.

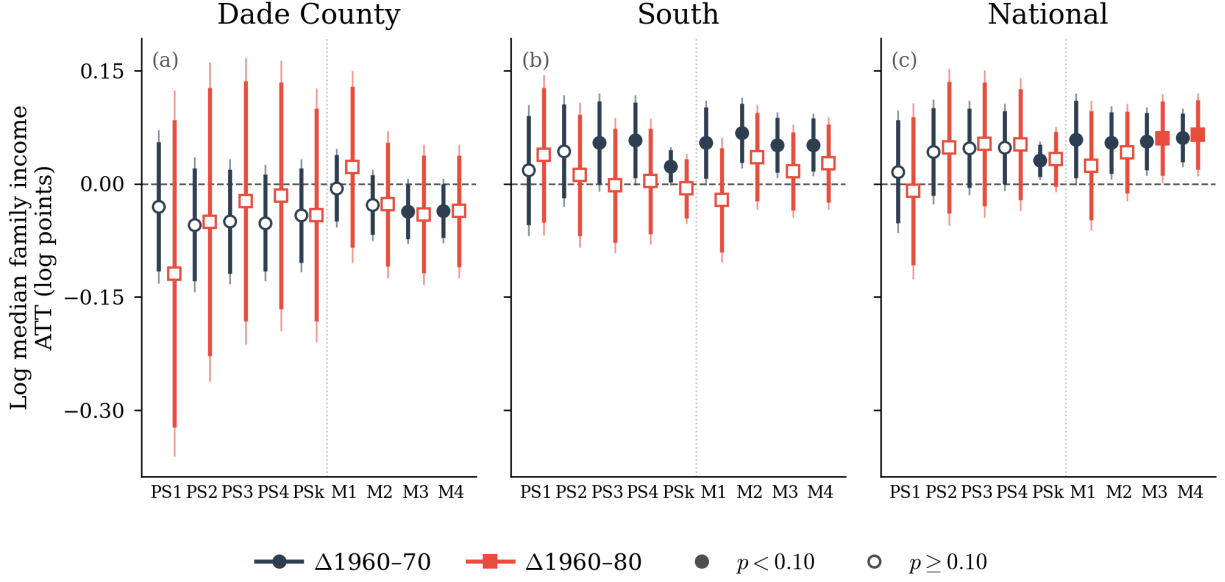


Figure 6: Log median family income: every matching estimator, by control pool. Estimates are average effects on the treated for the 1960–70 and 1960–80 changes. Marker fill indicates $p < 0.10$; thick whiskers are 90% confidence intervals and thin whiskers 95%. PS1–PS4 and PSk are `psmatch2` with a logit propensity score and common support (one to four nearest neighbours and a kernel), bootstrapped 250 replications; M1–M4 are `teffects nnmatch` on the Mahalanobis metric with Abadie–Imbens bias adjustment. No estimate is selected on significance. Treatment is ≥ 3 elite households; tracts holding one or two are dropped rather than used as controls.

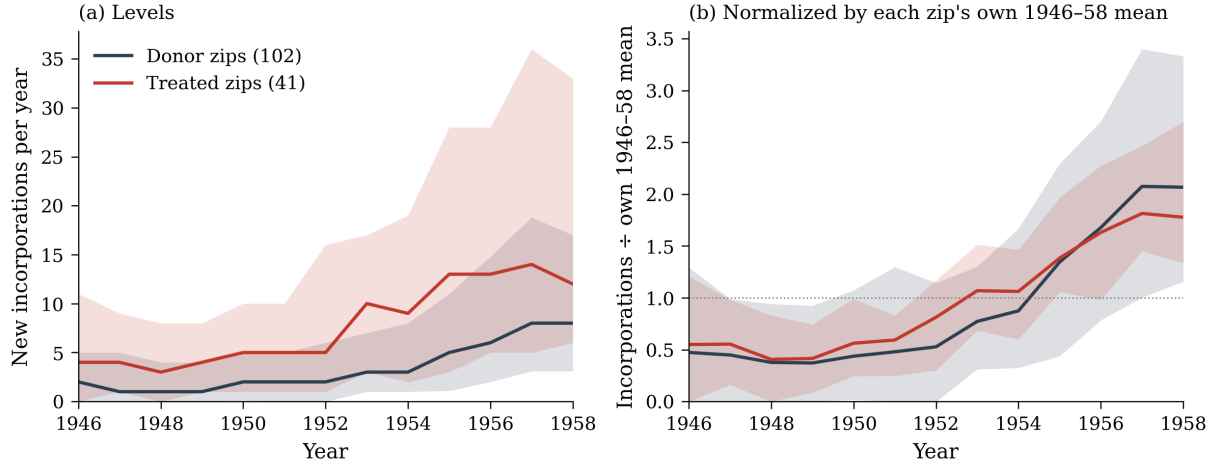


Figure 7: Treated and donor zip codes before treatment, 1946–58. (a) New incorporations per zip code per year. (b) Each zip code's incorporations divided by its own 1946–58 mean. Lines are medians across zip codes and shaded bands span the 10th to 90th percentiles. Treated: the 41 zip codes receiving at least three elite households. Donors: the 102 zip codes with no elite household and at least 26 filings between 1946 and 1958. Filings without a ZIP code in the registry are assigned one with the US Census geocoder.

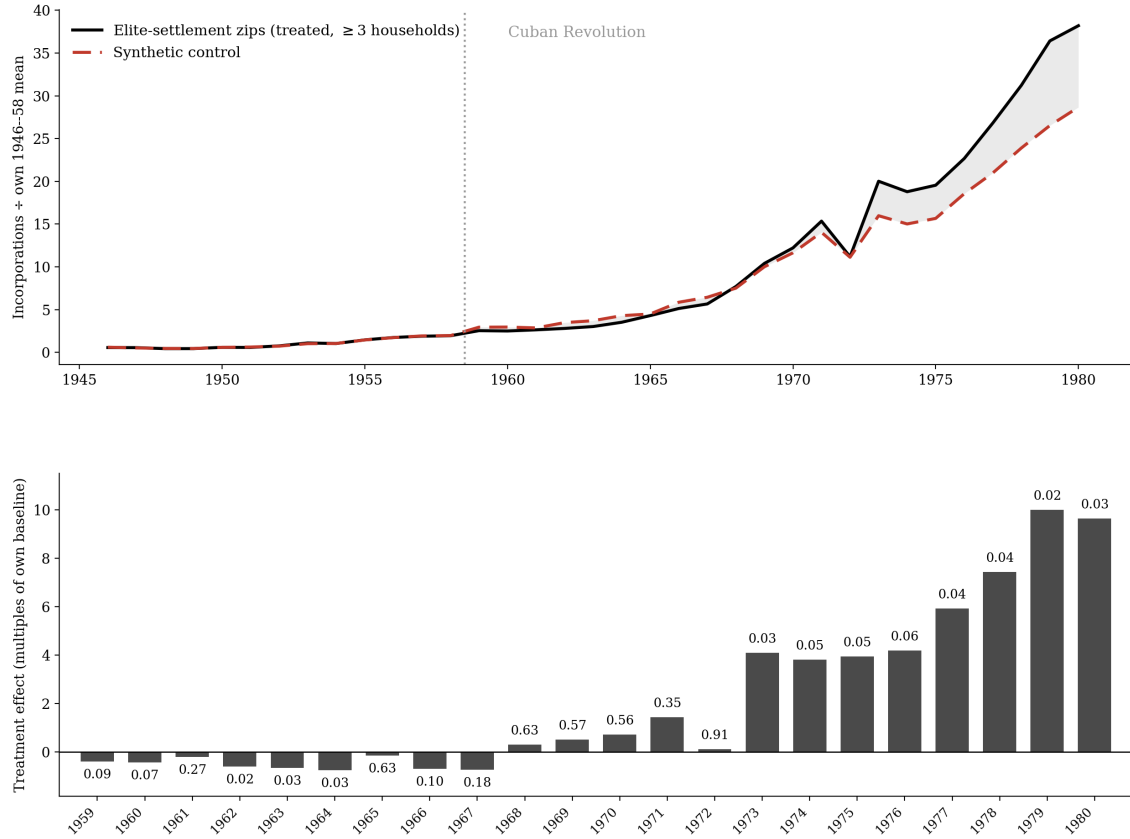


Figure 8: Synthetic control estimates of business formation, 1946–80. The outcome is each zip code’s new Florida incorporations divided by its own 1946–58 mean; filings without a ZIP code in the registry are assigned one with the US Census geocoder. Treated: the 41 zip codes receiving at least three elite households. Donors: the 102 zip codes with no elite household and at least 26 filings between 1946 and 1958. Top panel: mean treated and synthetic series. Bottom panel: effect by year with two-sided p -values from 100,000 placebo bundles, each the mean of 41 donor gaps drawn with replacement and scored exactly as the treated aggregate, $p = (\text{rank} - 1)/(N + 1)$.

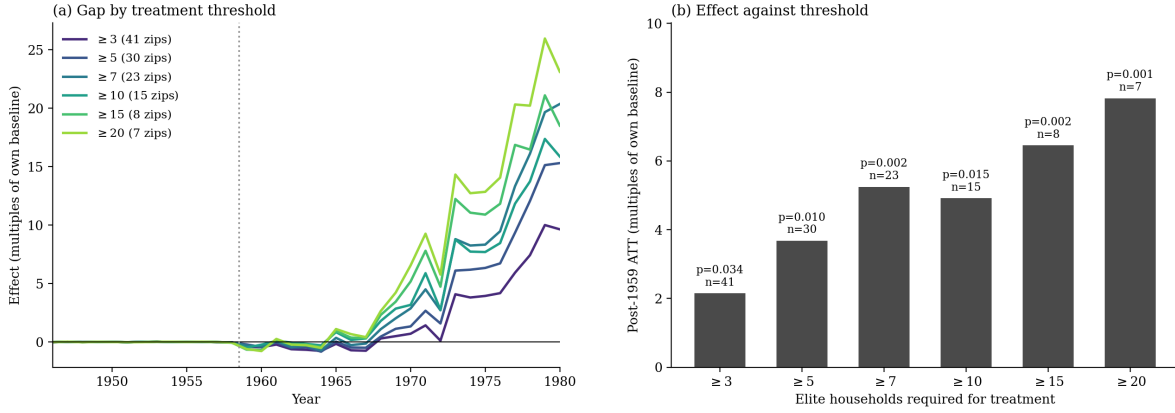


Figure 9: Dose-response. Every threshold reuses the same per-unit synthetic controls, fitted once against the same 102 donor zip codes, so raising it changes only which treated zips enter the average — not the estimator, not the donors, not any individual fit. Dose is not confounded with size: elite count correlates 0.11 with pre-1959 incorporations. Panel (b)’s p -values come from 100,000 placebo bundles of k donors, k matching the number of treated zips at that threshold.

10 Appendix

The matching results, outcome by outcome

Each table reports all nine matching estimators for one outcome, against each of the three control pools; Figures 4–6 plot the same estimates.

Table A1: Unemployment rate: the effect of elite settlement, $\Delta 1960$ –70

Matching Method	Dade		South		National	
	coef. 41–47	X^2 bal.	coef. 1,750–1,759	X^2 bal.	coef. 9,277–9,324	X^2 bal.
PS: 1 nearest	0.006 [0.51]	12.50 [0.09]	-0.015** [0.04]	7.79 [0.35]	-0.025*** [0.00]	16.35 [0.02]
PS: 2 nearest	0.006 [0.43]	5.16 [0.64]	-0.014** [0.02]	4.27 [0.75]	-0.020*** [0.00]	10.14 [0.18]
PS: 3 nearest	0.003 [0.66]	5.02 [0.66]	-0.013** [0.02]	3.75 [0.81]	-0.021*** [0.00]	14.13 [0.05]
PS: 4 nearest	0.005 [0.48]	3.21 [0.87]	-0.012** [0.02]	4.48 [0.72]	-0.020*** [0.00]	10.08 [0.18]
PS: normal kernel	0.006 [0.39]	1.72 [0.97]	-0.014*** [0.00]	35.67 [0.00]	-0.017*** [0.00]	52.24 [0.00]
Mahalanobis: 1 nearest	0.008** [0.03]	n.a.	0.002 [0.46]	n.a.	-0.019*** [0.00]	n.a.
Mahalanobis: 2 nearest	0.005* [0.10]	n.a.	-0.000 [0.93]	n.a.	-0.022*** [0.00]	n.a.
Mahalanobis: 3 nearest	0.005* [0.09]	n.a.	-0.001 [0.63]	n.a.	-0.022*** [0.00]	n.a.
Mahalanobis: 4 nearest	0.006* [0.07]	n.a.	-0.000 [0.89]	n.a.	-0.022*** [0.00]	n.a.

“PS” is propensity-score matching. p -values are in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Propensity-score standard errors are bootstrapped with 250 replications; Mahalanobis standard errors are bias-adjusted following Abadie and Imbens (2011). Control counts appear under each pool.

The X^2 columns report the chi-square statistic testing the null that the covariates are, on average, balanced between the treated tracts and their matched controls, computed over the matched sample. Balance is not available for the Mahalanobis estimator, which does not produce the matching weights the test requires.

Treated tracts are 46 throughout; tracts holding one or two elite households are excluded from the controls. Covariates are the five 1960 characteristics of Table 3 together with the outcome’s own 1950 and 1960 levels. Propensity-score matching is restricted to the region of common support; Mahalanobis matching uses all treated tracts.

Table A2: Unemployment rate: the effect of elite settlement, $\Delta 1960-80$

Matching Method	Dade		South		National	
	coef. 41–47	X^2 bal.	coef. 1,750–1,759	X^2 bal.	coef. 9,277–9,324	X^2 bal.
PS: 1 nearest	-0.013 [0.31]	6.82 [0.45]	-0.037*** [0.00]	16.20 [0.02]	-0.040*** [0.00]	3.88 [0.79]
PS: 2 nearest	-0.018* [0.10]	5.79 [0.56]	-0.029*** [0.00]	9.86 [0.20]	-0.027*** [0.00]	4.59 [0.71]
PS: 3 nearest	-0.022** [0.02]	5.85 [0.56]	-0.028*** [0.00]	7.76 [0.35]	-0.029*** [0.00]	5.45 [0.61]
PS: 4 nearest	-0.018* [0.05]	2.60 [0.92]	-0.027*** [0.00]	5.96 [0.54]	-0.027*** [0.00]	5.60 [0.59]
PS: normal kernel	-0.016 [0.11]	2.43 [0.93]	-0.028*** [0.00]	35.55 [0.00]	-0.034*** [0.00]	52.24 [0.00]
Mahalanobis: 1 nearest	-0.024*** [0.00]	n.a.	-0.012* [0.09]	n.a.	-0.029*** [0.00]	n.a.
Mahalanobis: 2 nearest	-0.020*** [0.00]	n.a.	-0.014** [0.03]	n.a.	-0.021*** [0.00]	n.a.
Mahalanobis: 3 nearest	-0.024*** [0.00]	n.a.	-0.014** [0.01]	n.a.	-0.023*** [0.00]	n.a.
Mahalanobis: 4 nearest	-0.023*** [0.00]	n.a.	-0.011** [0.04]	n.a.	-0.022*** [0.00]	n.a.

“PS” is propensity-score matching. p -values are in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Propensity-score standard errors are bootstrapped with 250 replications; Mahalanobis standard errors are bias-adjusted following Abadie and Imbens (2011). Control counts appear under each pool.

The X^2 columns report the chi-square statistic testing the null that the covariates are, on average, balanced between the treated tracts and their matched controls, computed over the matched sample. Balance is not available for the Mahalanobis estimator, which does not produce the matching weights the test requires.

Treated tracts are 46 throughout; tracts holding one or two elite households are excluded from the controls. Covariates are the five 1960 characteristics of Table 3 together with the outcome’s own 1950 and 1960 levels. Propensity-score matching is restricted to the region of common support; Mahalanobis matching uses all treated tracts.

Table A3: Employment/population: the effect of elite settlement, $\Delta 1960-70$

Matching Method	Dade		South		National	
	coef. 41-47	X^2 bal.	coef. 1,750-1,759	X^2 bal.	coef. 9,277-9,324	X^2 bal.
PS: 1 nearest	0.005 [0.88]	15.14 [0.03]	0.005 [0.75]	5.93 [0.55]	0.033* [0.08]	31.00 [0.00]
PS: 2 nearest	-0.012 [0.65]	1.58 [0.98]	0.009 [0.52]	2.92 [0.89]	0.035** [0.03]	18.76 [0.01]
PS: 3 nearest	-0.016 [0.49]	1.82 [0.97]	0.010 [0.43]	3.58 [0.83]	0.034** [0.03]	9.24 [0.24]
PS: 4 nearest	-0.011 [0.62]	0.99 [1.00]	0.007 [0.57]	4.16 [0.76]	0.033** [0.02]	8.00 [0.33]
PS: normal kernel	-0.007 [0.77]	0.55 [1.00]	0.007 [0.44]	38.14 [0.00]	0.010 [0.22]	n.a.
Mahalanobis: 1 nearest	0.081*** [0.00]	n.a.	0.002 [0.84]	n.a.	0.020** [0.02]	n.a.
Mahalanobis: 2 nearest	0.026* [0.07]	n.a.	0.001 [0.89]	n.a.	0.016* [0.06]	n.a.
Mahalanobis: 3 nearest	0.010 [0.43]	n.a.	0.005 [0.57]	n.a.	0.015* [0.06]	n.a.
Mahalanobis: 4 nearest	0.018 [0.12]	n.a.	0.005 [0.57]	n.a.	0.015* [0.09]	n.a.

“PS” is propensity-score matching. p -values are in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Propensity-score standard errors are bootstrapped with 250 replications; Mahalanobis standard errors are bias-adjusted following Abadie and Imbens (2011). Control counts appear under each pool.

The X^2 columns report the chi-square statistic testing the null that the covariates are, on average, balanced between the treated tracts and their matched controls, computed over the matched sample. Balance is not available for the Mahalanobis estimator, which does not produce the matching weights the test requires.

Treated tracts are 46 throughout; tracts holding one or two elite households are excluded from the controls. Covariates are the five 1960 characteristics of Table 3 together with the outcome’s own 1950 and 1960 levels. Propensity-score matching is restricted to the region of common support; Mahalanobis matching uses all treated tracts.

Table A4: Employment/population: the effect of elite settlement, $\Delta 1960-80$

Matching Method	Dade		South		National	
	coef. 41-47	X^2 bal.	coef. 1,750-1,759	X^2 bal.	coef. 9,277-9,324	X^2 bal.
PS: 1 nearest	-0.009 [0.90]	15.14 [0.03]	0.000 [0.99]	5.93 [0.55]	0.032 [0.22]	21.50 [0.00]
PS: 2 nearest	-0.010 [0.87]	1.58 [0.98]	-0.006 [0.75]	2.92 [0.89]	0.047** [0.04]	14.14 [0.05]
PS: 3 nearest	-0.006 [0.91]	1.82 [0.97]	-0.000 [0.99]	3.58 [0.83]	0.032 [0.12]	7.37 [0.39]
PS: 4 nearest	0.012 [0.80]	0.99 [1.00]	-0.003 [0.85]	4.16 [0.76]	0.026 [0.17]	7.60 [0.37]
PS: normal kernel	-0.004 [0.94]	0.55 [1.00]	-0.005 [0.64]	38.14 [0.00]	-0.004 [0.71]	n.a.
Mahalanobis: 1 nearest	0.079 [0.11]	n.a.	-0.002 [0.86]	n.a.	0.019* [0.08]	n.a.
Mahalanobis: 2 nearest	0.088** [0.02]	n.a.	-0.009 [0.46]	n.a.	0.011 [0.37]	n.a.
Mahalanobis: 3 nearest	0.134*** [0.00]	n.a.	-0.008 [0.50]	n.a.	0.012 [0.34]	n.a.
Mahalanobis: 4 nearest	0.109*** [0.01]	n.a.	-0.010 [0.34]	n.a.	0.010 [0.39]	n.a.

“PS” is propensity-score matching. p -values are in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Propensity-score standard errors are bootstrapped with 250 replications; Mahalanobis standard errors are bias-adjusted following Abadie and Imbens (2011). Control counts appear under each pool.

The X^2 columns report the chi-square statistic testing the null that the covariates are, on average, balanced between the treated tracts and their matched controls, computed over the matched sample. Balance is not available for the Mahalanobis estimator, which does not produce the matching weights the test requires.

Treated tracts are 46 throughout; tracts holding one or two elite households are excluded from the controls. Covariates are the five 1960 characteristics of Table 3 together with the outcome’s own 1950 and 1960 levels. Propensity-score matching is restricted to the region of common support; Mahalanobis matching uses all treated tracts.

Table A5: Median family income (log): the effect of elite settlement, $\Delta 1960-70$

Matching Method	Dade		South		National	
	coef. 41-47	X^2 bal.	coef. 1,750-1,759	X^2 bal.	coef. 9,277-9,324	X^2 bal.
PS: 1 nearest	-0.030 [0.56]	1.14 [0.98]	0.018 [0.69]	2.90 [0.82]	0.016 [0.70]	1.73 [0.94]
PS: 2 nearest	-0.054 [0.24]	1.81 [0.94]	0.043 [0.25]	0.84 [0.99]	0.042 [0.23]	3.98 [0.68]
PS: 3 nearest	-0.050 [0.24]	0.72 [0.99]	0.055* [0.10]	0.87 [0.99]	0.047 [0.14]	2.86 [0.83]
PS: 4 nearest	-0.052 [0.19]	0.92 [0.99]	0.058* [0.06]	1.04 [0.98]	0.048 [0.10]	2.00 [0.92]
PS: normal kernel	-0.042 [0.27]	0.66 [1.00]	0.023* [0.07]	29.29 [0.00]	0.031** [0.02]	52.39 [0.00]
Mahalanobis: 1 nearest	-0.006 [0.83]	n.a.	0.054* [0.06]	n.a.	0.059* [0.06]	n.a.
Mahalanobis: 2 nearest	-0.028 [0.25]	n.a.	0.067*** [0.00]	n.a.	0.054** [0.03]	n.a.
Mahalanobis: 3 nearest	-0.036* [0.10]	n.a.	0.051** [0.02]	n.a.	0.056** [0.01]	n.a.
Mahalanobis: 4 nearest	-0.036* [0.10]	n.a.	0.052** [0.02]	n.a.	0.061*** [0.00]	n.a.

“PS” is propensity-score matching. p -values are in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Propensity-score standard errors are bootstrapped with 250 replications; Mahalanobis standard errors are bias-adjusted following Abadie and Imbens (2011). Control counts appear under each pool.

The X^2 columns report the chi-square statistic testing the null that the covariates are, on average, balanced between the treated tracts and their matched controls, computed over the matched sample. Balance is not available for the Mahalanobis estimator, which does not produce the matching weights the test requires.

Treated tracts are 46 throughout; tracts holding one or two elite households are excluded from the controls. Covariates are the five 1960 characteristics of Table 3 together with the outcome’s own 1950 and 1960 levels. Propensity-score matching is restricted to the region of common support; Mahalanobis matching uses all treated tracts.

Table A6: Median family income (log): the effect of elite settlement, $\Delta 1960-80$

Matching Method	Dade		South		National	
	coef. 41–47	X^2 bal.	coef. 1,750–1,759	X^2 bal.	coef. 9,277–9,324	X^2 bal.
PS: 1 nearest	-0.119 [0.34]	6.10 [0.41]	0.038 [0.48]	2.88 [0.82]	-0.010 [0.87]	13.57 [0.03]
PS: 2 nearest	-0.050 [0.64]	1.03 [0.98]	0.012 [0.81]	6.01 [0.42]	0.048 [0.36]	4.12 [0.66]
PS: 3 nearest	-0.023 [0.81]	1.53 [0.96]	-0.002 [0.96]	7.31 [0.29]	0.053 [0.29]	5.54 [0.48]
PS: 4 nearest	-0.016 [0.86]	0.46 [1.00]	0.003 [0.94]	2.34 [0.89]	0.052 [0.24]	4.21 [0.65]
PS: normal kernel	-0.041 [0.63]	0.82 [0.99]	-0.006 [0.79]	29.30 [0.00]	0.032 [0.14]	52.40 [0.00]
Mahalanobis: 1 nearest	0.022 [0.73]	n.a.	-0.022 [0.61]	n.a.	0.024 [0.58]	n.a.
Mahalanobis: 2 nearest	-0.027 [0.58]	n.a.	0.035 [0.32]	n.a.	0.041 [0.21]	n.a.
Mahalanobis: 3 nearest	-0.041 [0.39]	n.a.	0.017 [0.59]	n.a.	0.060** [0.04]	n.a.
Mahalanobis: 4 nearest	-0.036 [0.42]	n.a.	0.027 [0.39]	n.a.	0.065** [0.02]	n.a.

“PS” is propensity-score matching. p -values are in brackets. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Propensity-score standard errors are bootstrapped with 250 replications; Mahalanobis standard errors are bias-adjusted following Abadie and Imbens (2011). Control counts appear under each pool.

The X^2 columns report the chi-square statistic testing the null that the covariates are, on average, balanced between the treated tracts and their matched controls, computed over the matched sample. Balance is not available for the Mahalanobis estimator, which does not produce the matching weights the test requires.

Treated tracts are 46 throughout; tracts holding one or two elite households are excluded from the controls. Covariates are the five 1960 characteristics of Table 3 together with the outcome’s own 1950 and 1960 levels. Propensity-score matching is restricted to the region of common support; Mahalanobis matching uses all treated tracts.

Robustness: a higher treatment dose

Table A7: Robustness: the tract design at higher treatment doses

Outcome	Dose	Dade	South	National
Panel A. $\Delta 1960-70$				
Unemployment rate	≥ 3	+0.006 (4/9)	-0.012* (5/9)	-0.021* (9/9)
	≥ 5	+0.006 (4/9)	-0.016* (7/9)	-0.019* (5/9)
	≥ 7	+0.009 (4/9)	-0.006 (3/9)	-0.024* (8/9)
Employment/population	≥ 3	+0.005 (2/9)	+0.005 (0/9)	+0.020* (8/9)
	≥ 5	+0.034 (2/9)	+0.008 (0/9)	+0.028* (6/9)
	≥ 7	+0.014 (2/9)	+0.026 (3/9)	+0.041* (6/9)
Median family income (log)	≥ 3	-0.036 (2/9)	+0.052* (7/9)	+0.048* (5/9)
	≥ 5	-0.040 (1/9)	+0.057 (4/9)	+0.043 (4/9)
	≥ 7	-0.043 (0/9)	+0.079 (4/9)	+0.095* (5/9)
Panel B. $\Delta 1960-80$				
Unemployment rate	≥ 3	-0.020* (7/9)	-0.027* (9/9)	-0.027* (9/9)
	≥ 5	-0.023* (5/9)	-0.025* (8/9)	-0.036* (9/9)
	≥ 7	-0.024 (4/9)	-0.026* (6/9)	-0.044* (9/9)
Employment/population	≥ 3	+0.012 (3/9)	-0.005 (0/9)	+0.019 (2/9)
	≥ 5	+0.048 (3/9)	+0.004 (0/9)	+0.022 (2/9)
	≥ 7	-0.007 (2/9)	+0.031 (2/9)	+0.052* (6/9)
Median family income (log)	≥ 3	-0.036 (0/9)	+0.012 (0/9)	+0.048 (2/9)
	≥ 5	-0.036 (1/9)	+0.021 (0/9)	+0.058 (3/9)
	≥ 7	-0.023 (1/9)	-0.004 (1/9)	+0.078 (4/9)

Median estimate across the nine matching estimators; number significant at $p < 0.10$ in parentheses. * marks a majority.

“Dose” is the number of elite households a tract must receive to count as treated. The adopted specification uses ≥ 3 , giving 46 treated tracts; ≥ 5 and ≥ 7 give 23 and 15.

Raising the dose shrinks the treated sample, so significance thins mechanically; the check is whether the sign and magnitude survive. For the 1960–80 change in the unemployment rate they do in every pool, and against the national pool the estimate grows with the dose, from -0.027 to -0.036 and -0.044.

Business formation: robustness

Table A8: Business formation: synthetic difference-in-differences

Estimator	ATT	s.e.	<i>t</i>	95% CI
Synthetic difference-in-differences	+30.49	9.39	+3.25	[+12.1, +48.9]

Outcome is new Florida incorporations per zip code per year, in levels, 1946–80, treatment 1959. Synthetic difference-in-differences follows Arkhangelsky et al. (2021), estimated with the `sdid` package of Clarke et al. (2023).

The outcome here is a firm count, while the main specification is normalised by each zip’s own pre-treatment mean. The two magnitudes are therefore not directly comparable.

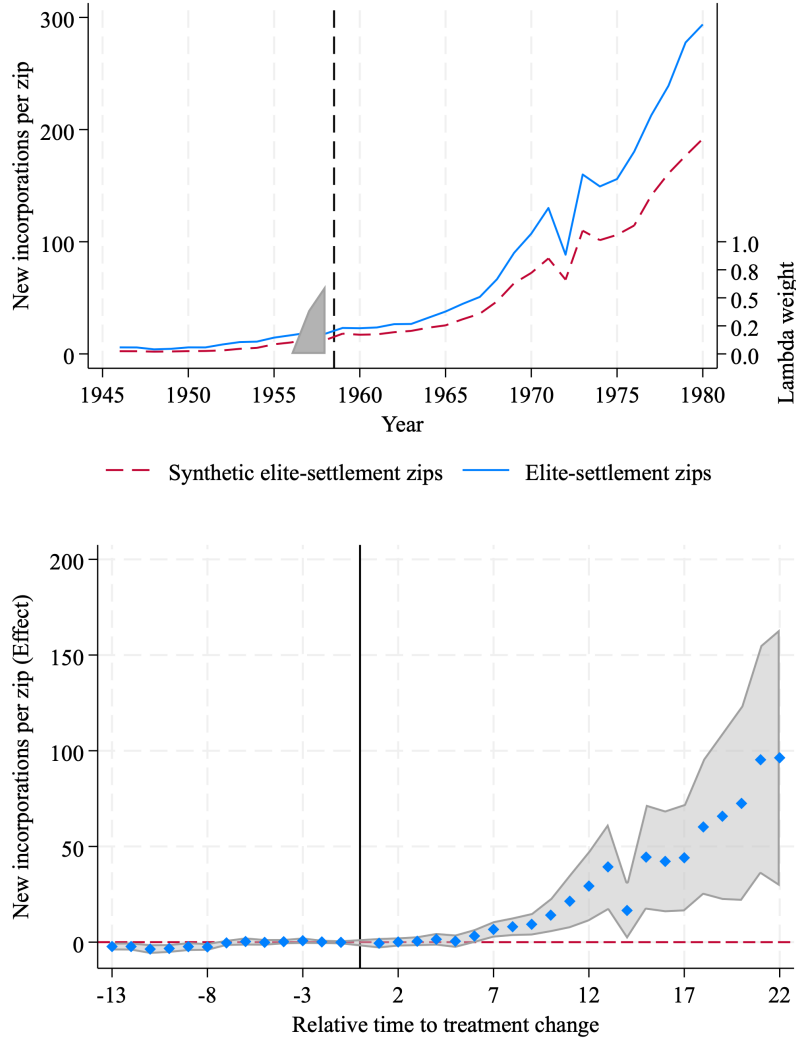


Figure A1: Synthetic difference-in-differences, new incorporations per zip code per year. Estimated with the `sdid` package of Clarke et al. (2023) following Arkhangelsky et al. (2021), outcome in levels, the 41 zip codes receiving at least three elite households against the same 102 donor zip codes used by the synthetic control, so only the estimator differs; filings without a ZIP code in the registry are assigned one with the US Census geocoder. Upper panel: treated and synthetic series, with the estimator's time weights λ shaded below. Lower panel: the `sdid_event` study in relative time, with 95% bootstrap confidence bands from 500 replications.